Reverse Chain Rule

Exam code: 9MA0

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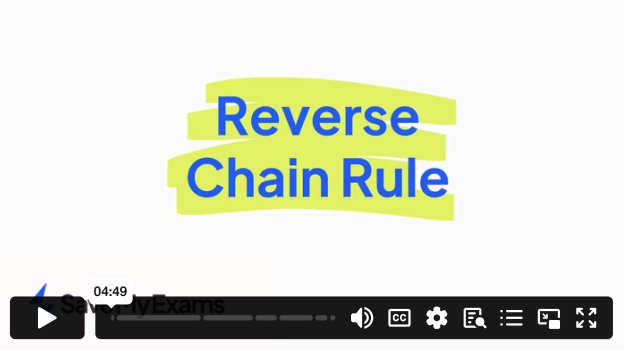

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Help me understand

Reverse chain rule

What is the chain rule?

- The Chain Rule is a way of differentiating two (or more) functions

FOR $y = f(u)$, WHERE $u = g(x)$

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$

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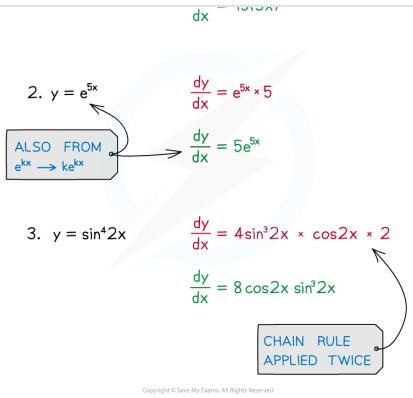
- In many simple cases the above formula/substitution is not needed

Stuck on something?

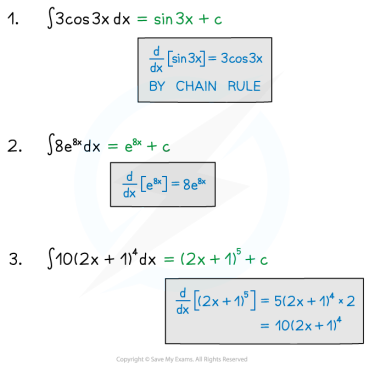
Ask questions and get help with exactly what you're revising.

Got it

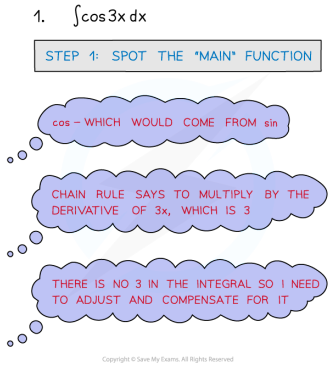


- The same can apply for the reverse – integration

How do I integrate using the reverse chain rule?



- In more awkward cases it can help to write the numbers in before integrating
- STEP 1: Spot the 'main' function
- STEP 2: 'Adjust' and 'compensate' any numbers/constants required in the integral
- STEP 3: Integrate and simplify



$= \frac{1}{3}\int 3\cos 3x \, dx$

3cos3x IS WHAT I'D EXPECT TO SEE FROM DIFFERENTIATING sin3x USING CHAIN RULE

STEP 3: INTEGRATE AND SIMPLIFY

$= \frac{1}{3}\sin 3x + c$

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Examiner Tips and Tricks

- If in doubt you can always use a substitution.
- Differentiation is easier than integration so if stuck try the opposite, eg. **sin** and **cos** are linked (remember that minus!) so if integrating a **sin** function, start by differentiating the corresponding **cos** function.
- Lastly, check your final answer by differentiating it.

Worked Example

Find the following integrals

(a) $\int \cos 3x \, dx$

(b) $\int 5e^{6x} \, dx$

(c) $\int (2x+5)^3 \, dx$

a) $\int \cos 3x \, dx$

STEP 1: cos IS THE MAIN FUNCTION

STEP 2: • "ADJUST": 3
• "COMPENSATE": $\frac{1}{3}$

$= \frac{1}{3}\int 3\cos 3x \, dx$

$= \frac{1}{3}\sin 3x + c$

STEP 3: 3cos3x IS THE DIFFERENTIAL OF sin3x BY CHAIN RULE

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